

AREKATLA NISHANTH CHOWDARY

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OBJECTIVE

Aspiring Image Processing and AI engineer with hands-on expertise in deep learning-based segmentation, computer vision, and medical image analysis. Experienced with CNNs including SAM (Segment Anything Model), OpenCV, and Python-based AI pipelines. Seeking to contribute to the development of oncology and surgical imaging platforms, including image enhancement, organ segmentation, and GPU-optimized deep learning systems.

EDUCATION

Amrita Vishwa Vidyapeetham <i>B.Tech in Computer Science Engineering (with AI) — CGPA: 7.23</i>	2023 – Present
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TECHNICAL SKILLS

Image Processing: Image enhancement, denoising, reconstruction, segmentation (classical + deep learning), OpenCV, Matlab
Deep Learning: CNNs, U-Net, ResNet, Vision Transformers, SAM (Segment Anything Model), Quantization, Unsloth fine-tuning, VLLMs, transfer learning
Programming: Python (primary), C, C++, Java, JavaScript
Frameworks & Tools: OpenCV, PyTorch, Jupyter Notebook, VS Code, Matlab, Git, Flask, Linux
AI/ML Concepts: Machine Learning, Deep Learning, NLP, Generative AI, RAG, IoT, GPU computing
Databases: MongoDB, MySQL, SQLite
Soft Skills: Communication, Team Collaboration, Problem Solving, Leadership

EXPERIENCE

Viswam.AI, IIIT Hyderabad — AI Developer Intern <i>On-Site</i>	Apr 2025 – Jul 2025
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- Worked hands-on with Segment Anything Model (SAM) — directly relevant to organ segmentation pipelines — alongside Quantization and Unsloth fine-tuning for deep learning model optimization.
- Built a custom parameter-efficient fine-tuned model demonstrating advanced segmentation and image understanding capabilities aligned with medical imaging requirements.
- Collaborated on cutting-edge AI research focused on model deployment and real-world performance optimization, contributing to scalable AI-driven imaging workflows.
- Gained hands-on exposure to VLLMs (Vision-Language Large Models) for image comprehension tasks relevant to oncology and surgical imaging analysis.

PROJECTS

Smart Traffic Management System YOLOv8, OpenCV, Python, IoT, Pandas, NumPy	Jun 2024
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- Developed a real-time vehicle detection and tracking pipeline using YOLOv8 and OpenCV, applying deep learning-based segmentation and object detection — core skills for organ segmentation in surgical imaging.
- Implemented frame-wise tracking, virtual zone-based decision logic, and real-time image analytics demonstrating end-to-end deep learning inference optimization.
- Reduced emergency vehicle response time by 35% through GPU-optimized real-time image processing, directly applicable to GPU optimization in medical imaging systems.

WARP — Waste Analysis Report Project Gemini API, Firebase, Flask	Jan 2025
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- Built an AI-powered environmental damage assessment platform generating automated reports for government intervention, integrating AI models into full-stack pipelines.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional (May 2025) — ID: 101666956OCI25GENAIPRO
- Oracle Cloud Infrastructure 2025 AI Foundations Associate (May 2025) — ID: 101666956OCI25AICFA
- REST API Intermediate (Jun 2025) — ID: 53aca2a32f9f
- Joy of Computing with Python — NPTEL (Dec 2024) — ID: NPTEL24CS113S755802494

ACHIEVEMENTS

Gold Medal — 10th Research Day, SRM University (2026)

- Secured the Gold Medal for the paper 'Quantum Enhanced Wafer Failure Prediction using QSVM and QRF on 4 and 8 Qubits' — demonstrating advanced research capability at the intersection of quantum computing and AI-driven predictive modeling.

Silver Medal — 9th Research Day, SRM University (2025)

- Represented Amrita Vishwa Vidyapeetham and secured the Silver Medal in the Image Processing track for the paper 'Emergency Vehicle Detection and Traffic Management using YOLOv8x' — demonstrating recognized expertise in image processing and deep learning.

Team Leader, Robotics Club — Amrita Vishwa Vidyapeetham (Jan 2025 – Present)

- Led the Robotics Club fostering innovation and collaboration at the intersection of robotics and AI — relevant experience for cross-functional collaboration in oncology/surgical product development.